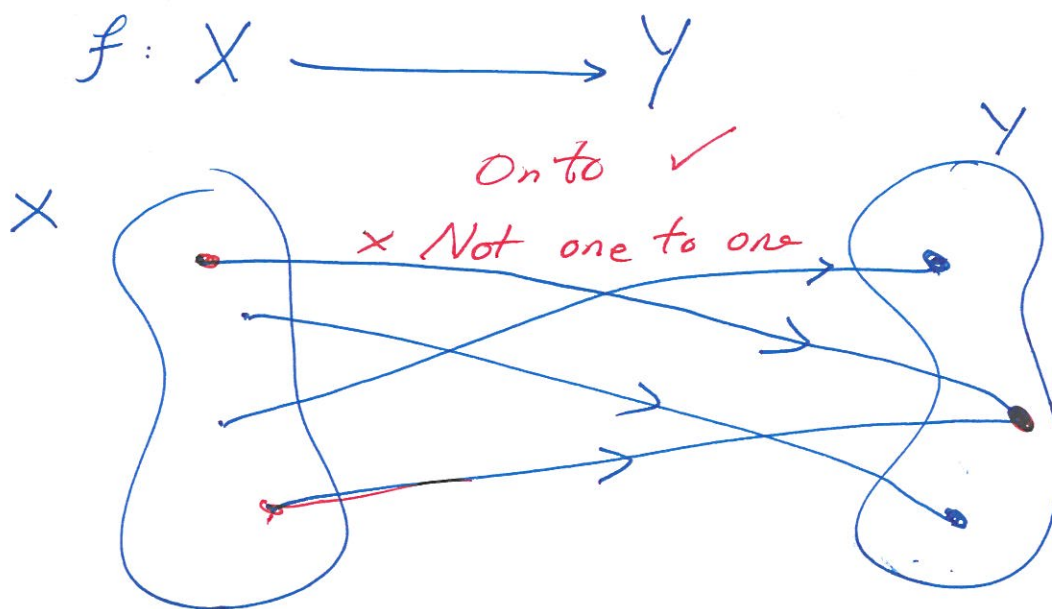


LECTURE 20: MATH1061/7861
TOPICS: ONE TO ONE, ONTO, AND INVERSE
FUNCTIONS
DATE: WEDNESDAY APRIL 15, 2026

Learning objectives:

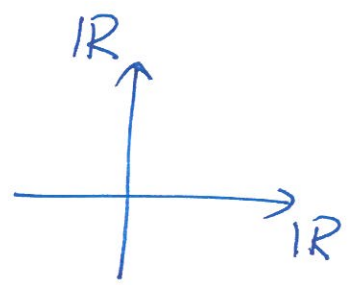
- (1) Learn how to prove a function is one to one, onto, or bijective.
- (2) Learn the inverse of a function.



Onto (Surjective) means that every element of Y is hit.

One to one (injective) means no two elements of X map to the same element of Y .

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\}$$



Q1: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x + y$.

(a) Is f one-to-one? Justify.

No. $f(1, 2) = f(2, 1) = 3$

(b) Is f onto? Justify.

Yes. Given a real number y ,

$$f(0, y) = y.$$

domain.

codomain

Q2: Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $g(x) = 2x + 3$.

(a) Is g injective?

Yes

Suppose. $g(x) = g(y)$ $x, y \in \mathbb{Z}$

$$\Rightarrow 2x + 3 = 2y + 3,$$

$$\Rightarrow \underline{\underline{x = y}}$$

(b) Is g surjective?

No.

We cannot get even numbers as outputs of g because $2x + 3$ is always odd.

$\nwarrow$ i.e. set of all outputs
 (c) What is the image of g ? This is $\mathbb{Z}^{\text{odd}}$

So thought of as function $\mathbb{Z} \rightarrow \mathbb{Z}^{\text{odd}}$
 g is injective & surjective.

(d) Determine the inverse of $g : \mathbb{Z} \rightarrow \mathbb{Z}^{\text{odd}}$.

$$\begin{array}{ccc}
 x & \xrightarrow{\quad} & 2x+3 \\
 \mathbb{Z} & \xrightarrow{\quad g \quad} & \mathbb{Z}^{\text{odd}}
 \end{array}$$

$$\mathbb{Z} \xleftarrow{\quad g^{-1} \quad} \mathbb{Z}^{\text{odd}}$$

$$\frac{x-3}{2} \xleftarrow{\quad} x$$

Q 3

Let A be a set with 10 elements
and B a set with 5 elements.

Can I have an injective function from $A \rightarrow B$?
No. Because A has more elements than B , there's bound
to be two elements in A which map
to same
element in B .
What about a surjective function?

